

Name Key

Total 12 points

Please show all your work.

1. (5 pts) The number of vacation days used by a sample of 15 employees in a recent year is given. Find the data set's first, second, and third quartiles, interquartile range and determine if there are any outliers.

6 9 0 12 9 6 7 5 11 4 5 22 8 9 13

0 4 5 5 6 6 7 8 9 9 9 11 12 13 22

$$IQR \times 1.5 = 6 \times 1.5 = 9$$

$$11 + 9 = 20$$

$$5 - 9 = -4$$

$$Q1 \underline{5} \quad Q2 \underline{8}$$

$$Q3 \underline{11} \quad IQR \underline{6}$$

$$\text{Outliers} \underline{22}$$

2. (5 pts) The probability that a person in the U.S. has type O+ blood is 38%.

a) Find the probability that a randomly chosen person in the U.S. does not have type O+.

$$100 - 38 = 62$$

$$\underline{62\%}$$

b) If four unrelated people in the U.S. are selected at random, find the probability that all four have O+.

$$(.38)^4 \approx .021$$

$$\underline{2.1\%}$$

c) If three unrelated people in the U.S. are selected at random, find the probability that at least one of the three has type O+.

$$P(\text{at least one has } O^+) = 1 - P(\text{no one has } O^+)$$

$$= 1 - (.62)^3 = .76$$

$$\underline{76\%}$$

3. (2 pts) Four people are selected at random. Find the probability that all four are born on a different day of the week.

$$\frac{7}{7} \times \frac{6}{7} \times \frac{5}{7} \times \frac{4}{7} = .35$$

$$\underline{35\%}$$